Solve 15 + 27 and express it clearly...

Conversation Transcript: Solve 15 + 27 and express it clearly...

1. User / Student / Teacher

Solve 15 + 27 and express it clearly step by step.

2. ClassWise AI (Class 10)

The sum of 15 and 27 is 42.

Step-by-Step Addition

To solve the addition problem $15 + 27$ using standard column addition, we align the numbers vertically by their place values (tens and units columns) and add them column by column from right to left.

$$

\begin{array}{r@{\quad}l}

1\ 5 & (\text{1 ten and 5 units}) \\

+ 2\ 7 & (\text{2 tens and 7 units}) \\

\hline

\end{array}

$$

Step 1: Add the units column

Add the digits in the units place (5 and 7):

$$

5 + 7 = 12

$$

Since 12 consists of 1 ten and 2 units, write down 2 in the units column of the answer and carry over 1 ten to the tens column.

Step 2: Add the tens column

Add the digits in the tens place (1 and 2) along with the carried-over 1 ten:

$$

1 \text{ (carry)} + 1 + 2 = 4

$$

Write down 4 in the tens column of the answer.

Final Result

Combining the tens and units gives:

$$

42

$$

3. User / Student / Teacher

How do you add decimals like 15.5 and 27.8?

4. ClassWise AI (Class 10)

The sum of 15.5 and 27.8 is 43.3.

Step-by-Step Decimal Addition

To add decimals, align the numbers vertically by their decimal points so that corresponding place values (tenths, units, tens) line up properly. Fill in any missing decimal places with zeros if necessary.

$$

\begin{array}{r@{\quad}l}

1\ 5 \cdot 5 & (\text{1 ten, 5 units, and 5 tenths}) \\

+ 2\ 7 \cdot 8 & (\text{2 tens, 7 units, and 8 tenths}) \\

\hline

\end{array}

$$

Step 1: Add the tenths column (rightmost column)

Add the digits in the tenths place (5 and 8):

$$

5 + 8 = 13

$$

Write down 3 in the tenths place of the answer and carry over 1 unit to the units column.

Step 2: Add the units column

Add the digits in the units place (5 and 7) along with the carried-over 1 unit:

$$

1 \text{ (carry)} + 5 + 7 = 13

$$

Write down 3 in the units place of the answer and carry over 1 ten to the tens column.

Step 3: Add the tens column

Add the digits in the tens place (1 and 2) along with the carried-over 1 ten:

$$

1 \text{ (carry)} + 1 + 2 = 4

$$

Write down 4 in the tens column. Place the decimal point directly in line with the decimal points above.

Final Result

Combining all the place values yields:

$$

43.3

$$

5. User / Student / Teacher

Create a simple mermaid flowchart of photosynthesis: Sunlight -> Chlorophyll -> Glucose

6. ClassWise AI (Class 10)

Photosynthesis is the biochemical process by which green plants convert light energy into chemical energy to synthesize glucose from carbon dioxide and water, using chlorophyll as the light-absorbing pigment.

Process Flowchart

The following diagram illustrates the pathway from sunlight and chlorophyll to the production of glucose:

flowchart LR

A[Sunlight] -->|Captured by| B[Chlorophyll]

B -->|Drives conversion of CO2 and H2O| C[Glucose]

Detailed Explanation of the Steps

1. Sunlight: Light energy from the sun is absorbed by plant tissues, serving as the primary energy source to drive the endothermic chemical reaction of photosynthesis.

2. Chlorophyll: This is the green pigment located within the chloroplasts of plant cells. Its role is to absorb light energy (primarily in the blue and red wavelengths) and initiate the light-dependent reactions.

3. Glucose: Through a series of complex reactions (the light-dependent reactions and the Calvin cycle), the trapped solar energy is used to convert carbon dioxide (CO_2) and water (H_2O) into glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), which serves as food and energy storage for the plant.

The overall chemical equation for photosynthesis can be expressed as:

$$

6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{Light Energy & Chlorophyll}}
\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2

$$